MILITARY SPECIFICATION

STATIC TUBE: M8

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of static tube entitled Static Tube: M8 which is used for the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8785809	- Static Probe.
8785810	- Forward Yoke.
8785812	- Nose, Body.
8785815	- Adjusting Screw.
8738268	- Nose Block.
8738301	- Mast, Static Tube.
8840076	- Static Tube.
8840113	- Proby Assembly.
8850097	- Container, Metal, Ammunition for Dummy Static Tube: XM16.
EL-8869680	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

FSC: 1135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Static tube shall comply with all requirements specified on the drawing (dwg) 8840076, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Leakage. The leakage rate of the Static tube shall not exceed 3.00 millimeters (mm) of mercury (Hg) per minute per 10 cu. inch volume or inversely proportional to other volume ratios when tested as specified in 4.4.1.

Test	Test Code No.
Maximum Leakage Rate 3.00 mm of Hg per min. per 10 cu in. of Volume	030001

3.7 Rate of flow. The orifices in the probe assembly after assembly in the Static tube shall return the system (8 cu. inch) to atmospheric pressure + zero -15 mm of Hg within 4 seconds from 520 ± 25 mm. Hg negative pressure when tested as specified in 4.4.2.

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Test	Test Code No.
Maximum Time 4 seconds	040001

3.8 Movement of probe.

3.8.1 Degree of movement. The tip of the probe shall have freedom of movement in all directions of 3.34 inches minimum radius (6.68 inches diameter) centered about the rotational axis shown on dwg 8840076. The maximum movement shall be when the mechanical stop on the vane and body assembly contacts the mast, Static tube. At this point the vane shall not contact the mast, Static tube. Tests shall be as specified in 4.4.3.1.

3.8.2 Resistance to movement. The maximum torque to displace the probe from the centerline to the 3.35 ± 0.01 inches minimum radius shall not exceed 20 inch ounces (oz.). The maximum torque to return the probe to the centerline shall not exceed 2.0 inch oz. Tests shall be as specified in 4.4.3.2.

Test	Test Code No.
Maximum Torque to Displace 20 inch-ounces	052001
Maximum Torque to Center 2.0 inch-ounces	052002

3.9 Probe balance. The Static tube shall be balanced within 0 to 0.75 in. oz. when tested in accordance with 4.4.4.

Test		Test Code No.
Torque to Displace	R1 equals	060001
	R2 equals	060002
	R3 equals	060003
	R4 equals	060004
The difference between any two values must be within 0 and 0.75 inch-ounces.		060005

3.10 Environmental testing. The Static tube shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The Static tube shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

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3.10.1 Temperature shock. The Static tube shall show no evidence of mechanical damage or deterioration after being subject to a sudden environmental change in temperature from -65 ± 5 degrees Fahrenheit (°F.) to 600 ± 10°F. when tested as specified in 4.4.5.1. After the thermoshock, the Static tube shall comply with the requirements of 3.6 and 3.8.2.

3.10.2 Operational vibration. The Static tube shall withstand a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affectiong functioning when tested as specified in 4.4.5.2. The total time of test shall be 210 seconds; 70 seconds in each of three mutually perpendicular axes, one of which shall be the longitudinal axis of the Static tube. The complex vibration test consists of the following five sections programmed continuously:

a. White Gaussian noise at a level of 22 g root mean square (RMS) band-limited between 15 and 1500 Hertz (Hz) for one second.
b. A 4 g RMS swept sine wave superimposed on 7.5 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. An 8 g RMS swept sine wave superimposed on 15 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 4 g RMS sine wave superimposed on 7.5 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 22 g RMS band-limited between 15 and 1500 Hz for one second.

Following the vibrations the Static tube shall comply with the requirements of 3.6 and 3.8.2.

3.10.3 Transportation vibration. The Static tube shall withstand vibration for 15 minutes in each of three mutually perpendicular axes, one of which shall be the longitudinal axis of the Static tube, at the frequencies and amplitudes specified below without evidence of damage or affecting functioning when tested as specified in 4.4.5.3. The frequency shall be varied uniformly from 8 to 500 to 8 Hz in 15 minutes. After being subjected to the specified vibration, the Static tube shall comply with the requirements of 3.6 and 3.8.2.

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Frequency range (Hz)	Cycle time	Constant acceleration or double amplitude
8-55-8	7-1/2 minutes	0.060 ± 0.006 inch
55-800-55	7-1/2 minutes	10 ± 1 g

3.10.4 Acceleration. The Static tube shall withstand a constant acceleration of 40 ± 2 gravity units (g) for ten seconds minimum in both directions of the longitudinal axis and 25 ± 1 g for ten seconds minimum in both directions of each of the other two mutually perpendicular axes without evidence of damage or affecting functioning when tested as specified in 4.4.5.4. After being subjected to the specified acceleration, the Static tube shall comply with the requirements of 3.6 and 3.8.2.

3.10.5 Shock. The Static tube shall show no evidence of damage when subjected to shock of 25 ± 2 g three times in each direction along the three mutually perpendicular axes, one of which shall be the longitudinal axis of the Static tube, for a total of 18 shocks when tested as specified in 4.4.5.5. Upon completion of the shocks, the Static tube shall comply with the requirements of 3.6 and 3.8.2.

3.11 Center of gravity. The center of gravity of the movable mass, less bellows, shall be as specified on drawing 8840076 when tested as specified in 4.4.6.

3.12 Test codes. The contractor shall provide data feedback on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contractor requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8840076) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Leakage (see 3.6)		X
3. Rate of flow (see 3.7)		X
4. Degree of movement (see 3.8.1)		X
5. Resistance to movement (see 3.8.2)		X
6. Probe balance (see 3.9)		X
7. Temperature shock (see 3.10.1)	X	
8. Operational vibration (see 3.10.2)	X	
9. Transportation vibration (see 3.10.3)	X	
10. Acceleration (see 3.10.4)	X	
11. Shock (see 3.10.5)	X	
12. Center of gravity (see 3.11)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Static probe (see dwg 8785809).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Wall thickness, minimum	Gage
	102. Location of orifice holes	Gage
	103. Size of orifice holes	Gage
	104. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.2 Nose block (see dwg 8838268).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Damaged shoulder, forward end	Visual
	102. Damaged bore	Visual
	103. Diameter of bore "A"	Gage
	104. Damaged surface (Sta. 8.125)	Visual

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105. Damaged locating diameter at station 8.125.....Visual
106. Damaged surface, rear end.....Visual
107. Length, Sta. 8.125 to rear surface of locking plate.....Gage
108. Pin hole location incorrect.....Gage
109. Pin hole size incorrect.....Gage
110. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.13).....Visual

4.3.2.1.3 Forward yoke (see dwg 8785810).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of external thread, undersize	Gage
102.	Major diameter of external thread, undersize	Gage
103.	Length of external thread undercut, maximum	Gage
104.	Damaged threads	Visual
105.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.4 Adjusting screw (see dwg 8785815).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread, undersize	Gage
102.	Major diameter of thread, undersize	Gage
103.	Length to thread, maximum	Gage

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104. Total length, minimum.....Gage
105. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.13).....Visual

4.3.2.1.5 Mast, static tube (see dwg 8838301).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Slot location incorrect.....Gage
102. Depth of slot insufficient.....Gage
103. Pitch diameter of thread, oversize.....Gage
104. Depth of thread, minimum.....Gage
105. Diameter damaged.....Visual
106. Shoulder surface damaged.....Visual
107. Insufficient braze on rear yoke-mast assembly with flux remaining.....Visual
108. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.13).....Visual
202. Serialization improper or missing (see 3.5).....Visual

4.3.2.1.6 Static tube (see dwg 8840076).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Planes of fins not mutually perpendicular.....Gage
102. Concentricity of Nose cone body at 11.90 location.....Gage
103. Concentricity of Static probe at 15.14 location.....Gage

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104. Concentricity of Static probe at 18.90 location.....Gage
105. Concentricity of fins at 8.00 location.....Gage
106. Static coupling outside diameter.....Gage
107. Diameter of packing groove.....Gage
108. Height of fin.....Gage
109. Length of fin.....Gage
110. Inability of forward yoke to move freely in relation to rear yoke.....Visual/Manual
111. Loose hardware.....Visual/Manual
112. Foreign matter or burrs in orifice holes.....Visual
113. Finish on nose of probe incorrect.....Visual
114. Improperly staked screws.....Visual
115. Scratched, nicked or gouged finish.....Visual
116. Damaged or missing packing (visible only).....Visual
117. Insufficient braze on Vane and Body assembly with remaining flux.....Visual
118. Incorrect or illegible marking including top vane relative to slot in mast.....Visual
119. Adjusting screw not seated properly.....Visual
120. Pin protruding above surface.....Visual
121. Width of packing groove.....Gage
122. Objective evidence relative to material missing or improper (see 3.1).....Visual

4.3.2.1.7 Nose, Body (see dwg 8785812).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of large thread, oversize	Gage
	102. Minor diameter of large thread, oversize	Gage

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103. Length to beginning of large thread, oversize.....Gage
104. Length to rear of thread, minimum.....Gage
105. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.13).....Visual

4.3.2.1.8 Probe assembly (see dwg 8840113).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Braze inadequate	Visual
	102. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.9 Container, Metal, Ammunition, For Static Tube: XM8 or Dummy Static Tube: XM16 (see dwg 8850097).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Packing material and packaging not in accordance with applicable dwg	Visual
	102. Marking not in accordance with applicable dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.13)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified.

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Test	Test Classification
a. Leakage	Major
b. Rate of flow	Major
c. Movement of probe	Major
d. Probe balance	Major
e. Transportation vibration	Major
f. Operational vibration	Major
g. Acceleration	Major
h. Temperature	Major

Of these tests e, f, g and h are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Leakage, rate of flow, movement of probe, and probe balance. Each Static tube shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.6, 3.7, 3.8 and 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration, operational vibration, acceleration, and temperature shock. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.5. If any item fails to comply with 3.10.1 through 3.10.4, the lot shall be rejected.

Table IV

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
3-50	2
51-150	3
151-300	4
301-600	5
601-1000	7
1001 and over	8

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-88869680).

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When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Leakage. Remove protective caps from front and rear orifice openings. The orifices in the probe shall be temporarily sealed. The static coupling shall be joined to a pressure sensing device, a positive shut off and a source of vacuum capable of obtaining 500mm. of mercury negative pressure. When the temperature of the part and testing equipment has stabilized at ambient and the pressure within the part and testing equipment has stabilized at 500 mm. of Hg. negative pressure, the positive shutoff shall be closed. The pressure sensing device shall be observed for 10 minutes, and the rate of leakage recorded. During the test, the probe shall be rotated 10 times at its maximum limit of travel.

4.4.2 Rate of flow. Remove protective caps from front and rear orifice openings. The orifices in the probe shall be temporarily sealed. The static coupling shall be connected to a pressure sensing device, a chamber of 8 $\pm$ 1 cubic inch volume, a positive stop and a source of 520 $\pm$ 25mm. of Hg. negative pressure. When the temperature of the part and testing equipment has stabilized at ambient and the pressure within the part and chamber has stabilized at 520 $\pm$ 25 mm. Hg. negative pressure, the positive shutoff shall be closed. The seal will then be removed from the probe orifices, and the time required for the system to return to atmospheric pressure + zero, -15 mm. of Hg. shall be recorded.

4.4.3 Movement of probe. Remove protective cap from front and rear orifice openings.

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4.4.3.1 Degree of movement. The static tube shall be mounted in a vertical position in a fixture which simulates the mounting characteristics on the missile. The fixture shall establish a zero centerline at the tip of the probe within 0.010 inches of the centerline of the mounting device. A right and left reference line shall be established at 3.34 distance, parallel to the zero centerline. At a minimum of four points, in line with each of the vanes on the assembly, the probe tip shall pass both reference lines. With the probe moved to the maximum displacement by the mechanical stop, the vanes shall not contact the mast static tube.

4.4.3.2 Resistance to movement. With the static tube mounted and positioned in the fixture referred to in 4.4.3.1 the torque required to move the probe tip to the 3.35 ± 0.01 displacement line shall be determined, at each of four positions located approximately midway between the vanes. At each position the torque to return the probe tip to the centerline shall also be determined. The maximum torque required to displace the probe and maximum torque required to return it to the zero position shall be recorded.

4.4.4 Probe balance. Remove protective caps from front and rear orifice openings. The static tube shall be mounted in a fixture which simulates the mounting characteristic on the missile. The fixture shall establish a zero centerline at the tip of the probe within 0.010 inch of the centerline of the mounting device. The fixture shall also establish a line that will indicate when the probe is displaced 12.5 ± 5° upward from the horizontal. This 12.5° shall be computed using mean dimensions of all components. The assembly shall be held vertical and the probe manually aligned to the zero mark on the fixture. The probe shall be displaced to the 12.5° mark and the torque required to so displace it recorded (R1). Place the fixture in a horizontal position and again deflect the probe to the 12.5° mark and record the torque (R2). After completing values R1 and R2, the assembly shall be rotated 90° and the test shall be repeated to obtain values R3 and R4. The difference between any two of the four readings must be within 0 to 0.75 in. oz.

4.4.5 Environmental testing (see 3.10).

4.4.5.1 Temperature shock. Remove protective caps from front and rear orifice openings. The static tube shall be placed in a cold chamber at -65 ± 5°F. and held there for 4 hours minimum after the termperature has stabilized (see 6.4). Then the static tube shall be transferred in not more than one minute to a hot chamber having a temperature of 600 ± 10°F and held in the hot chamber for 120 ± 20 seconds. When the static tube has returned to room ambient temperature after removal from hot chamber, it shall be tested for the requirements 3.6 and 3.8.2.

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4.4.5.2 Operational vibration. Remove protective caps from front and rear orifice openings. A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting an approved sponge rubber collar so as to contact the rear inner surface of the vane and body assembly and the mast. The static tube shall be mounted in a fixture, simulating actual mounting conditions, on a vibration machine and vibrated in three mutually perpendicualar axes using a standard noise or complex wave tape. The tape shall have the characteristics listed in Table V.

Table V

Operation vibration tape characteristics

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration)(1)	1	Set gain to give 2 g RMS
20-30	None	-	-
30-31	Noise (1)	11	22
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4
46-61	Noise (1) plus sinusoid (3)	8.5	17
	Noise alone	7.5	15
	Sinusoid alone	4.	8
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4
99-100	Noise (1)	11.	22

(1) Band limited (15-1500 Hz) white Gaussian noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

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For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptatic slope of at least 24 decibel (DB) per octave and 3 DB points at 15 and 1500 Hz shall be considered acceptable. Following the vibration the static tube shall be tested for 3.6 and 3.8.2.

4.4.5.2.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3DB or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system the standard noise tape is applied. The over-all gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.5.2 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g level obtained will conform to the levels listed in Table V.

4.4.5.3 Transportation vibration. The static tube shall be mounted in a fixture, simulating actual mounting conditions on a vibration machine and vibrated for 15 minutes in each of the three mutually perpendicular axes, at the frequencies and amplitudes specified in 3.10.3. The total test time for each static tube shall be tested for requirements 3.6 and 3.8.2.

4.4.5.4 Acceleration. Remove protective caps from front and rear orifice openings. The Static tube shall be mounted in a fixture simulating actual mounting conditions. A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting an approved sponge rubber collar so as to contact the rear inner surface of the vane and body assembly and the mast. The Static tube plus the fixture shall be subjected to acceleration as specified in paragraph 3.10.4. After the acceleration test, the static tube shall be tested for requirements 3.6 and 3.8.2.

4.4.5.5 Shock. Remove protective caps from front and rear orifice openings. The Static tube shall be mounted in a fixture simulating actual mounting conditions. A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting an approved sponge rubber collar so as to contact the rear inner surface of the vane and body assembly and the mast. The Static tube plus the fixture shall be subjected to the shocks specified in paragraph 3.10.5. Each of the 18 shocks shall have a total duration of 11 $\pm$ 2 milliseconds as measured between points which are 10 percent of the peaks. The time to peak shall be 5.5 $\pm$ 1 milliseconds. After this test, the Static tube shall be tested for requirements 3.6 and 3.8.2.

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4.4.6 Center of gravity. The center of gravity of the movable mass of the Static tube, less bellows and front protective cap, shall be determined by any moment method satisfactory to the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Static tube shall be packaged in accordance with dwg 8850097 using the specified moisture barrier.

5.1.2 Level C. The Static tube shall be packaged in accordance with dwg 8850097except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The Static tube shall be packed in accordance with dwg 8850097.

5.3 Marking. Marking shall be as specified in accordance with dwg 8850097.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.5, 3.12 and 4.3.4.1 will be delivered on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2062 (Rev 1) dated 2 April 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A175

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